

AMENDMENT NO. 1 MARCH 1986

TO

IS:10505-1983 CODE OF PRACTICE FOR CONSTRUCTION OF
FLOORS AND ROOFS USING PRECAST CONCRETE WAFFLE UNITS

(Page 5, clause 4.4) - Add the following sentence
at the end of the clause:

'Guidelines for design of floors/roofs using waffle
units is given at Appendix B.'

(Page 9, Appendix A) - Add the following new
Appendix B after A-2.1:

APPENDIX B
(Clause 4.4)

GUIDELINES FOR DESIGN OF FLOORS/ROOFS
USING WAFFLE UNITS

B-1. LOADS AND FORCES

B-1.1 Loads and forces shall be taken in accordance
with clause 17 of IS:456-1978*. In addition, slab
should be checked for incidental concentrated load
which is likely to occur during the construction.

B-2. ANALYSIS OF FLOORS/ROOFS

B-2.1 The floor/roof with waffle units up to a span of
6 metres having ribs of width not less than 100 mm
(excluding thickness of precast waffle ribs) spaced
at not more than 750 mm may be analysed as solid slab
spanning in two directions at right angles in accordan
with 23.4 of IS:456-1978* or as flat slab in accordanc
with 30 of IS:456-1978*.

*Code of practice for plain and reinforced concrete
(third revision).

B-2.2 The floor/roof with waffle units having span of more than 6 m and a rib spacing of more than 750 mm shall be designed as slab and grid beam system. The shear at the interface of precast and in-situ concrete shall be calculated and suitable shear keys/shear reinforcement shall be provided to avoid separation.

NOTE - When the span of waffle unit is more than 750 mm, the structural design of the unit should be made.

B-3. CONTROL OF DEFLECTION

B-3.1 If the system is analysed as per B-2.1, the deflection shall be restricted in accordance with 23 of IS:456-1978* and the depth of rib of the unit shall be considered as depth of slab.

B-3.2 If the system is analysed as per B-2.2, the deflection should be restricted in accordance with 22.1, 22.2 and 23 of IS:456-1978*.

B-4. DETAILING OF REINFORCEMENT

B-4.1 Shear stress in the ribs shall be calculated as per IS:456-1978*.

B-4.2 Detailing of the reinforcement shall be done in accordance with clause 25 of IS:456-1978*.

**Code of practice for plain and reinforced concrete (third revision).*

(BXC 32)